THE UNIVERSITY OF MELBOURNE

Student number

Semester 1 Final Subject Assessment, 2023

School of Mathematics and Statistics

MAST20026 Real Analysis

Reading time: 15 minutes — Writing time: 3 hours

This exam consists of 28 pages (including this page) with 11 questions and 100 total marks

Permitted Materials

- One double-sided handwritten A4 sheet
- No calculators are permitted.
- Any mobile phones or internet-enabled devices brought into the exam room must be **turned off** and placed on the floor under your table.

Instructions to Students

- Write your answers in the boxes provided on the exam. If you do not have enough space, tick the box near the bottom of page 28 and request a booklet from an invigilator—include the question number at the top of each page in the booklet.
- You must NOT remove this question paper, or any booklets provided to you, at the conclusion of the examination.

Instructions to Invigilators

- Students are to write their answers on the paper. They may request a booklet if they run out of space on the exam paper.
- This exam paper contains examinable material and must be collected, together with answer booklets if any, at the conclusion of the examination.

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Question 1 (7 marks)

(a) Let p and q be primitive statements. Use a truth table to prove the logical equivalence

$$[p \wedge (p \vee (\sim q))] \equiv p$$

Briefly explain how the workings you have done prove the logical equivalence.

(b) Let $f : [0, 1] \to \mathbb{R}$ be a function. Write the following statement in formal mathematical language:

f is bounded above on $[0, 1]$.

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Question 2 (9 marks)

Using a proof by induction, prove that $n^3 + 5n$ is divisible by 3 for all $n \in \mathbb{N}$.

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Question 3 (9 marks)

Let U be a set. Let A, B, and C be subsets of U.

(a) Write a formal proof of the following statement:

$$A \setminus (B \cup C) \subseteq (A \setminus B) \cup (A \setminus C)$$

(b) Is it true that

$$A \setminus (B \cup C) = (A \setminus B) \cup (A \setminus C)\,?$$

Justify your answer using a formal proof or a specific counter-example.

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Question 4 (5 marks)

Listed below are the Axioms of the Real Numbers.

O1. $\forall x, y \in \mathbb{R}$ exactly one of the following is true: $x < y$, $y < x$, or $x = y$.
O2. $\forall x, y, z \in \mathbb{R}$ if $x < y$ and $y < z$ then $x < z$.
A1. $\forall x, y \in \mathbb{R} \quad x + y \in \mathbb{R}$
A2. $\forall x, y \in \mathbb{R} \quad x + y = y + x$
A3. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$
A4. $\exists 0 \in \mathbb{R} \quad \forall x \in \mathbb{R} \quad x + 0 = x$
A5. $\forall x \in \mathbb{R} \quad \exists -x \in \mathbb{R} \quad x + (-x) = 0$
M1. $\forall x, y \in \mathbb{R} \quad xy \in \mathbb{R}$
M2. $\forall x, y \in \mathbb{R} \quad xy = yx$
M3. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$
M4. $\exists 1 \in \mathbb{R} \setminus \{0\} \quad \forall x \in \mathbb{R} \quad 1x = x$
M5. $\forall x \in \mathbb{R} \setminus \{0\} \quad \exists 1/x \in \mathbb{R} \quad x \cdot 1/x = 1$
D. $\forall x, y, z \in \mathbb{R} \quad x(y + z) = xy + xz$
OA. $\forall x, y, z \in \mathbb{R}$ if $x < y$ then $x + z < y + z$
OM. $\forall x, y \in \mathbb{R}$ if $x > 0$ and $y > 0$ then $xy > 0$
C. Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Using only the Axioms of the Real Numbers (and rules of logic), write a formal proof showing that

$$(\forall x, y \in \mathbb{R}) \quad -(x + y) = (-x) + (-y)$$

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Note. *Here and for the remainder of the exam, you may write informal proofs, and you may assume that algebra and inequalities with the real numbers behave the way you expect.*

Question 5 (11 marks)

Consider the set

$$A = \left\{ 5 + \frac{2}{n} \;\middle|\; n \in \mathbb{N}^+ \right\}.$$

(a) Show that 5 is a lower bound for A in $\mathbb{R}$.

(b) Show that 5 is the infimum of A in $\mathbb{R}$.

(c) What are the set of limit points of A? (You do not need to justify your answer with a proof.)

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Question 6 (12 marks)

Consider the sequence (f_n) defined by

$$f_n = \frac{n}{3n-1}$$

for each $n \in \mathbb{N}^+$.

(a) Use the $\epsilon - M$ definition of convergence to prove that the sequence (f_n) converges to $1/3$.

(b) Is the sequence (f_n) Cauchy? Briefly explain.

(c) Let (g_n) be the sequence defined by $g_n = (-1)^n f_n$ for each $n \in \mathbb{N}^+$. Using Part (a) and the theorems from lecture, show that (g_n) diverges.

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Question 7 (10 marks)

Let $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^2 - 5 & \text{if } x \geq 2 \\ -x & \text{if } x < 2 \end{cases}$$

for each $x \in \mathbb{R}$.

(a) Use the $\epsilon - \delta$ definition of limits to show that

$$\lim_{x \to 3} f(x) = 4.$$

(b) Use the $\epsilon - \delta$ definition of continuity to show that f is discontinuous at $x = 2$.

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Question 8 (11 marks)

Answer the following questions using theorems from lecture.

(a) Let $f : [0, 1] \to \mathbb{R}$ be a function which is continuous on $[0, 1]$. Assume that $f(x) > 0$ for all $x \in [0, 1]$.

Show that there exists a real number $\alpha > 0$ such that $f(x) \geq \alpha$ for all $x \in [0, 1]$.

(b) Let $f : [0, 1] \to \mathbb{R}$ be a continuous function such that $f(x) \in \mathbb{Q}$ for all $x \in [0, 1]$. Must f be a constant function? Justify your answer with a proof or specific counter-example.

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Question 9 (5 marks)

Let $f : [-1, 1] \to \mathbb{R}$ defined by

$$f(x) = \frac{x^2}{3x^2 + 1}$$

for each $x \in \mathbb{R}$. Without computing the integral explicitly, show that

$$\frac{1}{7} \leq \int_{-1}^{1} \frac{x^2}{3x^2 + 1}\, dx \leq \frac{11}{28}.$$

(Hint: Consider the partition $\{-1, -1/2, 1/2, 1\}$.)

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Note. *For Questions 10 and 11, you may assume without $\epsilon - M$ proofs that limits behave the way you expect.*

Question 10 (12 marks)

(a) For each of the following infinite series, use an appropriate test to prove that the series converges or diverges.

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3+8}$

(ii) $\displaystyle\sum_{n=1}^{\infty} \frac{3^{2n}}{4^{n+1}}$

(b) Find the radius of convergence and the interval of convergence for the power series

$$\sum_{n=1}^{\infty} \frac{x^n}{2^n\, n^2}.$$

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Note. *As stated above, for Question 11 you may assume without $\epsilon - M$ proofs that limits behave the way you expect.*

Question 11 (9 marks)

Let $f : \mathbb{R} \to \mathbb{R}$ be a function which is infinitely differentiable, i.e. the n-th derivative $f^{(n)}$ of f exists and is differentiable on $\mathbb{R}$ for all $n \in \mathbb{N}^+$.

(a) Show that if there exists a real numbers $M_0 > 0$ and $M > 0$ such that

$$|f^{(n)}(x)| \leq n!\, M_0\, M^n$$

for all $x \in \mathbb{R}$ and for all $n \in \mathbb{N}$, then there exists $\rho > 0$ such that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}\, x^n$$

for each $x \in (-\rho, \rho)$.

(b) Show that if there exists $\rho > 0$ such that the Taylor series of f converges absolutely to $f(x)$ at each $x \in (-\rho, \rho)$, then there exists a real numbers $M_0 > 0$ and $M > 0$ such that

$$|f^{(n)}(0)| \leq n!\, M_0\, M^n$$

for all $n \in \mathbb{N}$.

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End of Exam — Total Available Marks = 100

You must tick this box if you have used extra booklets ☐